

Question

The purple double salt crystal **A** dissolves in water to form a yellow solution. Adding a colorless weakly acidic solution (solute is **B**) to the solution of **A** causes the color to fade to colorless, producing **C**. **B** and **A** contain the same cation, and the point group of the anion of **B** is $D_{\infty h}$. Adjusting the solution of **A** to alkaline, adding excess sodium hypochlorite for oxidation, and then reacting with $\text{Ba}(\text{OH})_2$ produces **D**, which is also purple. Adding an alkaline solution of sodium tetraphenylborate to the solution of **B** produces a white precipitate **E**. The molecular weight of **A** rounded down is 503. **A** ~ **E** refer to the complete chemical formulas of the species, and these species can exist with strong stability under normal temperature and pressure.

How many of the following descriptions are correct?

- The most common crystal form of **A** is cubic crystal system
- The molecular weight of **E** is greater than 400
- The cation of **B** is composed of multiple atoms
- **D** has poor water solubility
- The yellow color of **A** dissolving in water is due to the ionization of a Brønsted acid
- The molecular weight of **C** rounded down is a prime number

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

Answer

Correct Answer: **D**

Detailed Explanation

Sodium tetraphenylborate is often used to identify K^+ ions, but it is also effective for NH_4^+ . Therefore, the cation in **A** and **B** is most likely K^+ , but NH_4^+ etc. cannot be excluded.

CHECKPOINT

1 PTS

Sodium tetraphenylborate is often used to identify K^+ , the cation in **B** is most likely K^+ , **E** might be $KB(C_6H_5)_4$

The purple solid **A** turns yellow after dissolving, and after being oxidized by the strong oxidant sodium hypochlorite under alkaline conditions, **D** is obtained, which is still purple. Possible elements include iodine, manganese, iron, etc.

Under alkaline conditions, iodine will be oxidized to periodate by sodium hypochlorite, both of which are colorless, so iodine can be excluded.

CHECKPOINT

0.5 PTS

Periodate is colorless, excluding iodine

Manganese in stable valence states at normal temperature and pressure rarely has yellow solutions, so it can be excluded.

CHECKPOINT

1 PTS

Yellow and stable manganese-containing solutions are rare, excluding manganese

$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is purple, and hydrolysis occurs after dissolution, turning yellow. Fe^{3+} is oxidized to ferrate FeO_4^{2-} under alkaline conditions, forming a precipitate with Ba^{2+} . All are consistent with the question.

CHECKPOINT

1 PTS

$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is purple

CHECKPOINT

1 PTS

Fe^{3+} is oxidized to purple FeO_4^{2-} under alkaline conditions

CHECKPOINT

1 PTS

D is BaFeO_4

Common crystals containing $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ include potassium (or ammonium) ferric sulfate dodecahydrate, ferric nitrate nonahydrate, etc. And the question requires that **A** is a double salt, calculate the molecular weight of $\text{KFe}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ as $39.10 + 55.85 + 2 \times (32.07 + 4 \times 16.00) + 12 \times (2 \times 1.008 + 16) = 503.282$.

CHECKPOINT

1 PTS

Calculate the molecular weight of $\text{KFe}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ as 503. **A** is $\text{KFe}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$

F^- can decolorize the Fe^{3+} solution to generate FeF_6^{2-} , and the fluorine-containing weakly acidic $D_{\infty h}$ point group anion is HF_2^- .

CHECKPOINT

1 PTS

B is KHF_2 . **C** is K_3FeF_6

Option	Judgment
The most common crystal form of A is cubic crystal system	Correct
The molecular weight of E is greater than 400	Incorrect
The cation of B is composed of multiple atoms	Incorrect
The water solubility of D is poor	Correct
Dissolving A in water turns yellow accompanied by the ionization of Brønsted acid	Correct
The molecular weight of C rounded down is a prime number	Incorrect

CHECKPOINT

1 PTS

3 items are correct, the answer is D

